

Hunters Hill High School
Mathematics Extension 1
Trial Examination, 2016



Hunters Hill
High School

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- The marks for each question are shown on the paper
- Show all necessary working in questions 11-14.

Total Marks: 70

Section I Pages 3-6
10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II Pages 7-12
60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Section I**10 marks Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1. The coefficient of x^4 in $(2x - 3)^{13}$ is

(A) $-\binom{13}{4}2^43^9$

(B) $-\binom{13}{4}2^93^4$

(C) $\binom{13}{9}2^43^9$

(D) $\binom{13}{9}2^93^4$

2. What is the domain and range of $y = \sin^{-1}\left(\frac{2x}{5}\right)$?

(A) Domain: $-1 \leq x \leq 1$; Range: $-\pi \leq y \leq \pi$

(B) Domain: $-1 \leq x \leq 1$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(C) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(D) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\pi \leq y \leq \pi$

3. A particle moves in a straight line. Its position at any time, t , is given by

$$x = 3 \cos 2t + 4 \sin 2t.$$

The acceleration in terms of x is:

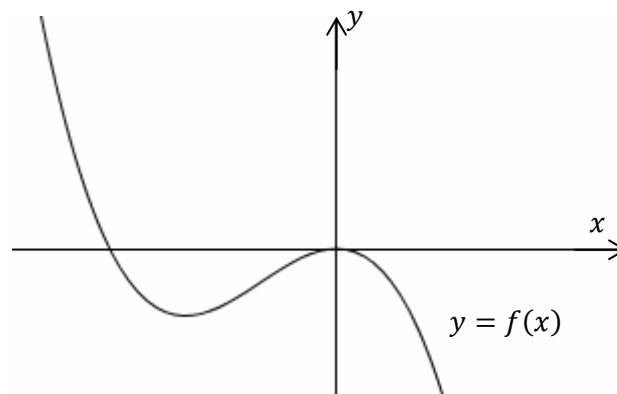
(A) $\ddot{x} = -3x$

(B) $\ddot{x} = -4x$

(C) $\ddot{x} = -16x^2$

(D) $\ddot{x} = -6 \cos 2x + 8 \sin 2x$

4. The diagram below shows the graph of a cubic function $y = f(x)$.



Which is a possible equation of this function?

- (A) $f(x) = -x(x - 2)(x + 2)$
(B) $f(x) = x^2(x - 2)$
(C) $f(x) = -x^2(x + 2)$
(D) $f(x) = -x^2(x - 2)$
5. What is the equation of the tangent at the point $(4p, 2p^2)$ on the parabola $x^2 = 8y$?
- (A) $y = px - p^2$
(B) $x + py = 2p + p^3$
(C) $x + py = 4p + p^3$
(D) $y = px - 2p^2$
6. The equation(s) of the horizontal asymptote(s) to the curve $y = \frac{x^2 + 1}{x^2 - 1}$ is/are:
- (A) $y = 0$
(B) $x = \pm 1$
(C) $y = 1$
(D) $x = 1$ only

7. Which of the following statements is FALSE?

- (A) $\cos^{-1}(-\theta) = -\cos^{-1} \theta$
- (B) $\sin^{-1}(-\theta) = -\sin^{-1} \theta$
- (C) $\tan^{-1}(-\theta) = -\tan^{-1} \theta$
- (D) $\cos^{-1}(-\theta) = \pi - \cos^{-1} \theta$

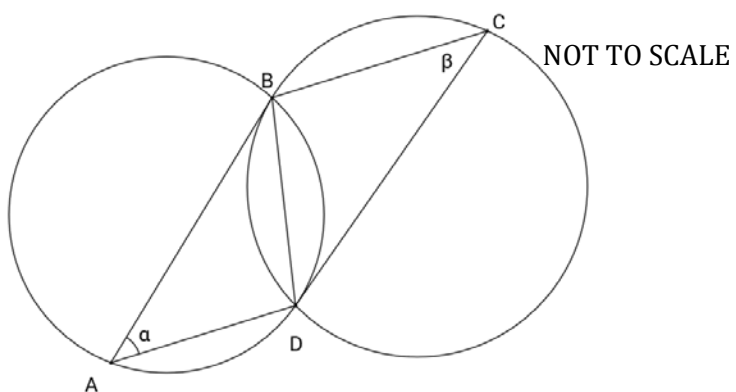
8. Three Mathematics study guides, four Mathematics textbooks and five exercise books are randomly placed along a bookshelf. What is the probability that the Mathematics textbooks are all next to each other?

- (A) $\frac{4!}{12!}$
- (B) $\frac{9!}{12!}$
- (C) $\frac{4! 3! 5!}{12!}$
- (D) $\frac{4! 9!}{12!}$

9. Water pours into a cylindrical tank of radius 50 cm at a rate of 1.5 L/s. The rate at which the height of the water in the tank is changing, with respect to time, is:

- (A) $\frac{3}{5000\pi} \text{ cm s}^{-1}$
- (B) $15\pi \text{ cm s}^{-1}$
- (C) $\frac{3}{\pi} \text{ cm s}^{-1}$
- (D) $\frac{3}{5\pi} \text{ cm s}^{-1}$

10. In the diagram below, AB is a tangent to the circle BCD , and CD is a tangent to the circle ABD . $\angle BAD = \alpha$ and $\angle BCD = \beta$.



Which of the following statements is true?

- (A) $\triangle ABD \equiv \triangle BDC$
- (B) $ABCD$ is a cyclic quadrilateral
- (C) $\triangle ABD \parallel \triangle BDC$
- (D) $AB \parallel CD$

Section II**60 marks****Attempt Questions 11–14****Allow about 1 hour and 45 minutes for this section**

Begin each question on a NEW SHEET of paper.

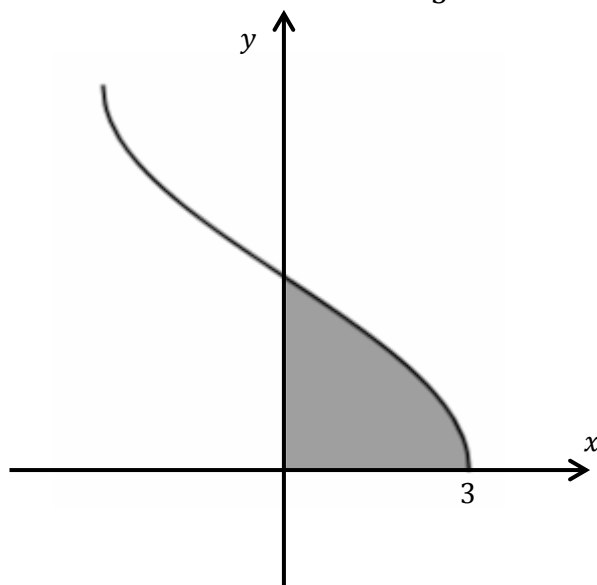
In questions 11 – 14 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- a. Let $A = (-1, 4)$ and $B = (5, -5)$. Find the coordinates of the point P which divides the interval AB in a ratio of 1: 2. 2
- b. A family consists of 8 people, including three young children.
- i. In how many ways can the family be arranged around a table so that the children sit together? 2
- ii. What is the probability that if randomly allocated a seat at the table, the children all sit together? 2
- c. Solve the inequality 2
- $$\frac{1}{|x - 1|} > \frac{1}{2}$$
- d. Differentiate $e^{\cos x} \ln x$ 2

Question 11 continued on next page

- e. The sketch shows the graph of the curve $y = 2 \cos^{-1} \frac{x}{3}$.

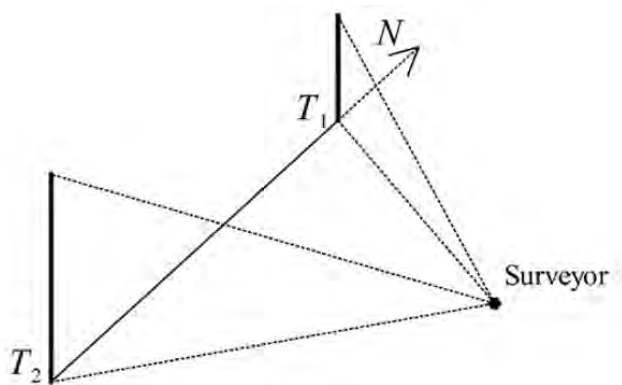


- | | | |
|------|---|---|
| i. | Find the y-intercept. | 1 |
| ii. | State the domain and range of the function. | 2 |
| iii. | Calculate the area of the function across the domain $0 \leq x \leq 3$ (the shaded region). | 2 |

End of Question 11

Question 13 (15 marks) Begin a SEPARATE sheet of paper.

- a. The velocity of a particle is given as $v^2 = 24 - 6x - 3x^2$.
- Find the particle's greatest displacement. 1
 - Hence, find the acceleration of the particle at its greatest displacement from origin. 2
- b. Find by division of polynomials, the remainder when $x^2 + 4$ is divided by $x - 3$. 2
- c. Use mathematical induction to prove that $3^n + 7^{n+1}$ is divisible by 4 for all integers $n \geq 1$. 3
- d. Evaluate $\int_0^{\frac{\pi}{4}} \sin^2 2x \, dx$ 3
- e. Two towers T_1 and T_2 have heights of h and $2h$ respectively. Tower T_2 is due south of T_1 . The bearing of tower T_1 from a surveyor is 292° , whilst the bearing of T_2 from the surveyor is 232° .
The angle of elevation from the surveyor to the top of tower T_1 is 30° and the angle of elevation from the surveyor to the top of tower T_2 is 60° .



Show that the distance d between the two towers is given by 4

$$d = \frac{\sqrt{21}h}{3} \text{ metres.}$$

End of Question 13

Question 14 (15 marks) Begin a SEPARATE sheet of paper.

- a. $f(x) = \sin^2 x - x + 1$ has a zero near $x_1 = \frac{\pi}{2}$. 2

Use one application of Newton's method to obtain another approximation, x_2 , to this zero.

- b. Use the substitution $u = \tan x$ to find an expression for 3

$$\int \frac{\sec^2 x}{\tan^2 x + 3} dx$$

- c. The binomial theorem is

$$c_0 + c_1x + c_2x^2 + \cdots + c_nx^n = (1 + x)^n$$

where $c_k = \binom{n}{k}$

- i. Show that 2

$$\frac{c_0x^2}{1.2} + \frac{c_1x^3}{2.3} + \frac{c_2x^4}{3.4} + \cdots + \frac{c_nx^{n+2}}{(n+1)(n+2)} = \frac{(1+x)^{n+2}}{(n+1)(n+2)}$$

- ii. Hence, find the following sum, writing your answer as a simple expression in terms of n . 1

$$\frac{c_0}{1.2} - \frac{c_1}{2.3} + \frac{c_2}{3.4} + \cdots + (-1)^n \frac{c_n}{(n+1)(n+2)}$$

Question 14 continues on next page.

- d. A fire hose is at ground level on a horizontal plane. Water is projected from the hose. The angle of projection, θ , is allowed to vary. The speed of the water as it leaves the hose, v metres per second, remains constant.

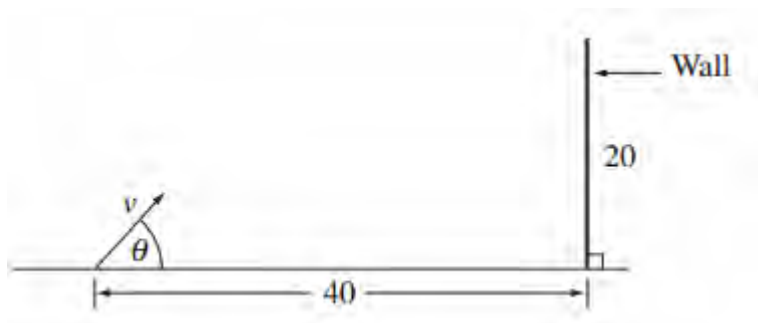
You may assume that if the origin is taken to be the point of projection, the path of the water is given by the parametric equations

$$\begin{aligned}x &= vt \cos \theta \\y &= vt \sin \theta - \frac{1}{2}gt^2\end{aligned}$$

where $g \text{ ms}^{-2}$ is the acceleration due to gravity. (Do NOT prove this)

- i. Show that the water returns to ground level at a distance $\frac{v^2 \sin 2\theta}{g}$ metres from the point of projection. 2

This fire hose is now aimed at a 20 metre high thin wall from a point of projection at ground level 40 metres from the base of the wall. It is known that when the angle θ is 15° , the water just reaches the base of the wall.

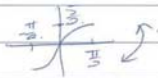


- ii. Show that $v^2 = 80g$. 1
- iii. Show that the Cartesian equation of the path of the water is given by 2
- $$y = x \tan \theta - \frac{x^2 \sec^2 \theta}{160}.$$
- iv. Show that the water just clears the top of the wall if 2
- $$\tan^2 \theta - 4 \tan \theta + 3 = 0.$$

End of paper

2016 TRIAL - MAX - SOLUTIONS

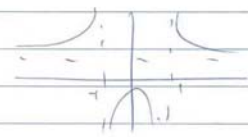
1. $\binom{13}{4} 2^4 (-3)^7 = -\binom{13}{4} 2^4 3^7$ (A)

2. $y = \frac{5}{2} \sin x$  (C)

3. $x = 3 \cos 2t + 4 \sin 2t$
 $\dot{x} = -6 \sin 2t + 8 \cos 2t$
 $\ddot{x} = -12 \cos 2t - 16 \sin 2t$
 $= -4(x)$ (B)

4. (C)

5. $\frac{dy}{dx} = \frac{x}{4} = \frac{4p}{4} = p$
 $y - 2p^2 = p(x - 4p)$
 $y = px - 2p^2$ (D)

6.  (C)

7. (A)

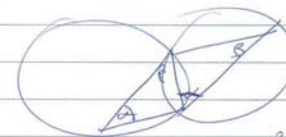
8. total = 12!
 ways to group 4 together = 4!9! (D)

9. $\frac{dV}{dt} = 1.5$ $V = \pi r^2 h$
 $= 2500\pi h$
 $h = \frac{V}{2500\pi}$

$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$
 $= \frac{1}{2500\pi} \cdot \frac{3}{2}$
 $= \frac{3}{5000\pi}$

(A)

10.



equilateral

(C)

Q11. c) $A=(-1,4), B(5,-5)$

2.

$$\begin{aligned} x &= \frac{1(5)+2(-1)}{1+2} & y &= \frac{1(4)+2(-5)}{1+2} \\ &= \frac{3}{3} & &= \frac{-6}{3} \\ &= 1 & &= -2 \end{aligned}$$

$\therefore P=(1,-2)$

b. i.



$$5! \cdot 3! = 120 \times 6 = 720$$

2.

ii.

$$\frac{5! \cdot 3!}{7!} = \frac{6}{8 \cdot 7} = \frac{1}{7}$$

2.

c.

$$\frac{1}{|x-1|} > \frac{1}{2}$$

$$|x-1| < 2$$

$$-2 < x-1 < 2$$

$$\therefore -1 < x < 3, x \neq 1$$

2.

d. $\frac{d}{dx} e^{\cos x} \ln x = e^{\cos x} \cdot \frac{1}{x} + (-\sin x) e^{\cos x} \cdot \ln x$

2.

e. i. $y = \pi$ at $x=0$

1.

$$\cos \frac{y}{2} = 0$$

$$\frac{y}{2} = \frac{\pi}{2}$$

$$y = \pi$$

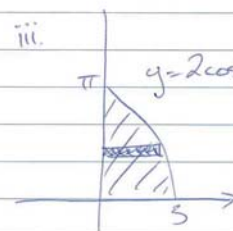


ii. domain: $-3 \leq x \leq 3$

2.

range: $0 \leq y \leq 2\pi$

iii.



2.

$$V = \int_0^{\pi} x \, dy$$

$$= 3 \int_0^{\pi} \cos \frac{y}{2} \, dy$$

$$= 3 \left[2 \sin \frac{y}{2} \right]_0^{\pi}$$

$$= 6 \left(\sin \frac{\pi}{2} - \sin \frac{0}{2} \right)$$

$$= 6(1-0)$$

$$= 6 \text{ units}^2$$

Q12

a. $x^3 - 2x^2 + 5x - 1 = 0$

i. $2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma)$
 $= 2\left(-\frac{-2}{1}\right)$
 $= 4.$

ii. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
 $= 2^2 - 2(5)$
 $= -6$

b. EQUATION 5V, 3C.

~~$5P_4 - 5P_4$~~
 ~~$= 1560.$~~
 ~~$5 \cdot 3 \cdot 6 \cdot 5 \times 4! = 10800$~~

c. PQ: $y = \frac{1}{2}(p+q)x - apq$

i) chord passes through $(0, a)$

$\therefore a = \frac{1}{2}(p+q) \cdot 0 - apq$

$a = -apq$
 $pq = -1.$

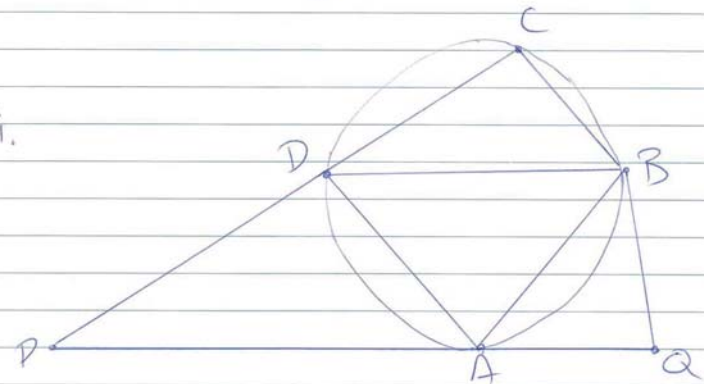
ii

$x = -apq(p+q)$
 $= -a(-1)(p+q)$
 $p+q = \frac{x}{a}.$

$y = a(p^2 + pq + q^2 + 2)$
 $= a(p^2 + 2pq + q^2 + 2 - pq)$
 $= a((p+q)^2 + 2 - pq)$
 $= a\left(\left(\frac{x}{a}\right)^2 + 2 - (-1)\right)$
 $= a\left(\frac{x^2}{a^2} + 3\right)$
 $x^2 = a(y - 3a)$

d.

i.



ii. $AP^2 = PD \cdot PC$ (Intersecting Tangent and Secant)
 $= 5 \cdot (5+7)$
 $= 60$
 $AP = 2\sqrt{15}$ units

iii. $\angle ADB = \angle BAQ$ (angle in alternate segment)
 $= \alpha$
 $\angle DBA = \angle BAQ$ (alternate angles in parallel lines)
 $= \alpha$

$\angle DAB + \angle ADB + \angle DBA = 180^\circ$ (angle sum of triangle)
 $\angle DAB + \alpha + \alpha = 180^\circ$
 $\angle DAB = 180 - 2\alpha$

$\angle BCD + \angle DAB = 180$ (opposite angles of cyclic quadrilateral)
 $\therefore \angle BCD = 180 - (180 - 2\alpha)$
 $= 2\alpha$

iv. $BQ = AQ$ (tangents to a point)
 $\angle ABQ = \angle BAQ$ (equal angles of isosceles triangle)
 $= \alpha$
 $\angle AQB + \alpha + \alpha = 180^\circ$ (angle sum of triangle)
 $\therefore \angle AQB = 180 - 2\alpha$

$\angle AQB + \angle BCD = 180 - 2\alpha + 2\alpha$
 $= 180^\circ$

$\therefore PQBC$ is a cyclic quadrilateral (opposite angles equal)

Q13 a. $v^2 = 24 - 6x - 3x^2$
 i. greatest displacement when $v=0$ 1.

$24 - 6x - 3x^2 = 0$
 $x^2 + 2x + 1 = 8 + 1$
 $(x+1)^2 = 9$
 $x+1 = \pm 3$
 $x = -1 \pm 3$
 $= -4, 2$

greatest displacement is 4 units

ii. $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ 2.
 $= \frac{d}{dx} \left(\frac{1}{2} (24 - 6x - 3x^2) \right)$
 $= \frac{1}{2} (-6 - 6x)$
 $= -3 - 3x$

at $x = -4$,
 $a = -3 - 3(-4)$
 $= 9 \text{ units/s}^2$

b. $\frac{x+3}{x-3}$ 2.
 $x-3 \overline{) x^2 + 4}$
 $\underline{x^2 - 3x}$
 $3x + 4$
 $\underline{3x - 9}$
 13
 remainder is 13.

c. Prove $3^n + 7^{n+1}$ is divisible by 4, for $n \geq 1$.

prove true for $n=1$

$$3^1 + 7^{1+1} = 3 + 49 = 52 \text{ which is divisible by 4.}$$

assume true for $n=k$

$$\text{i.e. } 3^k + 7^{k+1} = 4Q \text{ where } Q \text{ is an integer.}$$

$$3^k = 4Q - 7^{k+1}$$

prove true for $n=k+1$

$$\begin{aligned} 3^{k+1} + 7^{(k+1)+1} &= 3 \cdot 3^k + 7^{k+1} \cdot 7 \\ &= 3(4Q - 7^{k+1}) + 7 \cdot 7^{k+1} \\ &= 4 \cdot 3Q - 3 \cdot 7^{k+1} + 7 \cdot 7^{k+1} \\ &= 4 \cdot 3Q + 4 \cdot 7^{k+1} \\ &= 4(3Q + 7^{k+1}) \end{aligned}$$

which is divisible by 4.

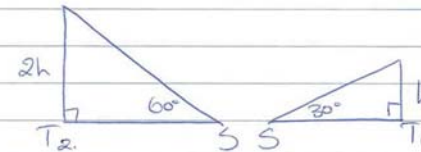
Hence, by induction $3^n + 7^{n+1}$ is divisible by 4 for $n \geq 1$.

d. $\int_0^{\frac{\pi}{4}} \sin^2 2x \, dx$ $1 - 2\sin^2 x = \cos 2x$ 3.

$$\begin{aligned} &= \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \cos 4x) \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin 4 \cdot \frac{\pi}{4} - \left(0 - \frac{1}{4} \sin 4(0) \right) \right] \\ &= \frac{1}{8} (\pi - 0) = \frac{\pi}{8} \end{aligned}$$

e.

4.

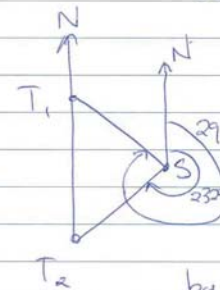
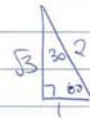


$$\tan 60^\circ = \frac{2h}{ST_2}$$

$$ST_2 = \frac{2h}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{h}{ST_1}$$

$$ST_1 = \sqrt{3}h$$



$$\angle T_1, ST_2 = 292 - 232 = 60^\circ$$

by cosine rule.

$$\begin{aligned} (T_1, T_2)^2 &= \left(\frac{2h}{\sqrt{3}} \right)^2 + (\sqrt{3}h)^2 - 2 \left(\frac{2h}{\sqrt{3}} \right) (\sqrt{3}h) \cos 60^\circ \\ &= \frac{4h^2}{3} + 3h^2 - 4h^2 \cdot \frac{1}{2} \\ &= \frac{4h^2 + 9h^2 - 6h^2}{3} \\ &= \frac{7h^2}{3} \\ T_1, T_2 &= \sqrt{\frac{7h^2}{3}} \\ &= \sqrt{\frac{7}{3}} h \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{21}h}{3} \text{ metres} \end{aligned}$$

Q14. a. $f(x) = \sin^2 x - x + 1$ $x_1 = \frac{\pi}{2}$ 2
 $f'(x) = 2 \sin x \cos x - 1$

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= \frac{\pi}{2} - \frac{\sin^2 \frac{\pi}{2} - \frac{\pi}{2} + 1}{2 \sin \frac{\pi}{2} \cos \frac{\pi}{2} - 1} \\ &= \frac{\pi}{2} - \frac{1^2 - \frac{\pi}{2} + 1}{2(1)(0) - 1} \\ &= \frac{\pi}{2} + (2 - \frac{\pi}{2}) \\ &= 2. \end{aligned}$$

b. $u = \tan x$ 3
 $du = \sec^2 x dx$

$$\begin{aligned} \int \frac{\sec^2 x}{\tan^2 x + 3} dx &= \int \frac{du}{u^2 + 3} \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} + C \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\tan x}{\sqrt{3}} \right) + C \end{aligned}$$

c. i. $C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n = (1+x)^n$ 2
 integrating:

$$\frac{C_0 x}{1} + \frac{C_1 x^2}{2} + \frac{C_2 x^3}{3} + \dots + \frac{C_n x^{n+1}}{n+1} = \frac{(1+x)^{n+1}}{n+1}$$

integrating:

$$\frac{C_0 x^2}{1 \cdot 2} + \frac{C_1 x^3}{2 \cdot 3} + \frac{C_2 x^4}{3 \cdot 4} + \dots + \frac{C_n x^{n+2}}{(n+1)(n+2)} = \frac{(1+x)^{n+2}}{(n+1)(n+2)}$$

ii. let $x = -1$

$$\frac{C_0 (-1)^2}{1 \cdot 2} + \frac{C_1 (-1)^3}{2 \cdot 3} + \frac{C_2 (-1)^4}{3 \cdot 4} + \dots + \frac{C_n (-1)^{n+2}}{(n+1)(n+2)} = \frac{(1-1)^{n+2}}{(n+1)(n+2)}$$

$$\frac{C_0}{1 \cdot 2} - \frac{C_1}{2 \cdot 3} + \frac{C_2}{3 \cdot 4} - \dots + \frac{(-1)^{n+2} C_n}{(n+1)(n+2)} = 0$$

d. $x = vt \cos \theta$ -I
 $y = vt \sin \theta - \frac{1}{2} g t^2$ -II

i. water is at ground level at $y=0$ 2

$$vt \sin \theta - \frac{1}{2} g t^2 = 0$$

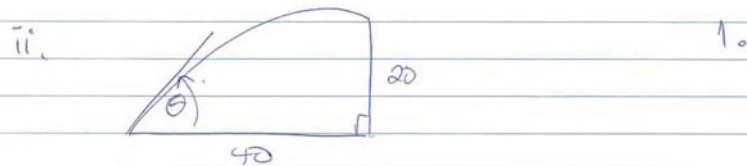
$$t(v \sin \theta - \frac{1}{2} g t) = 0$$

$$t=0 \text{ or } \frac{1}{2} g t = v \sin \theta$$

$$t = \frac{2v \sin \theta}{g}$$

substitute into II

$$\begin{aligned} x &= v \left(\frac{2u \sin \theta}{g} \right) \cos \theta \\ &= \frac{v^2}{g} 2 \sin \theta \cos \theta \\ &= \frac{v^2 \sin 2\theta}{g} \end{aligned}$$



when $\theta = 15^\circ$, $x = 40\text{m}$.

$$\therefore 40 = \frac{v^2 \sin 2(15^\circ)}{g}$$

$$v^2 = \frac{40g}{\sin 30^\circ}$$

$$= \frac{40g}{\frac{1}{2}}$$

$$\therefore v^2 = 80g$$



iii. from I

$$t = \frac{x}{v \cos \theta}$$

2.

Subst into II

$$\begin{aligned} y &= x \tan \theta - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2 \\ &= x \tan \theta - \frac{1}{2} g \frac{x^2}{v^2 \cos^2 \theta} \end{aligned}$$

$$= x \tan \theta - \frac{1}{2} g \frac{x^2 \sec^2 \theta}{80g}$$

$$\therefore y = x \tan \theta - \frac{x^2 \sec^2 \theta}{160}$$

iv. when $x = 40$, $y = 20$ water just clears the top. 2.

$$20 = 40 \tan \theta - \frac{40^2 \sec^2 \theta}{160}$$

$$= 40 \tan \theta - 10 (1 + \tan^2 \theta)$$

$$2 = 4 \tan \theta - 1 - \tan^2 \theta$$

$$\therefore \tan^2 \theta - 4 \tan \theta + 3 = 0$$